
A Phone Samples Space, Not Time

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Abstract

A location publisher is usually treated as a periodic sensor. It is not one. It arms an interval and a displacement together and fires whichever comes first, so above $v = D/I$ it steps off a fixed distance instead of a fixed time. We verify $\min(I, D/v)$ on three independent iPhones reporting to ThingsBoard on EC2, 20,201 fixes over 312 device-hours, and test it by changing D : the third handset ran 200 m instead of 50 m, and its step moved to 204 m and its crossover to 12 km/h as predicted. We then invert the law: arrival times alone recover speed to 15.0% median error against 6.1% for an estimator that reads the coordinates from the same instants, and label stop, walk and drive at 90.2%, 94.0% and 89.2%.

1 The law

OwnTracks in Move mode arms two conditions at once. `locatorInterval` is a timer; `locatorDisplacement` is a distance filter. A report goes out on either, so the gap between reports is

$$g(v) = \min\left(I, \frac{D}{v}\right), \quad v^* = D/I, \quad (1)$$

and with the lab's $I = 60$ s and $D = 50$ m the regimes exchange at $v^* = 3$ km/h. Below it the phone is a clock; above it, a ruler. Walking is 4 to 5 km/h, so a walk-only capture sits on the boundary and returns a clean 60 s period that hides the rule producing it, which is what our first capture did. Figures 1–4 separate the branches and price them. That the policy and not the hardware sets the shape is visible in Figure 10: the same iPhone also runs the Home Assistant companion app, which publishes on a timer, and over the same hours its step varies by three decades while its gap holds near 301 s, the mirror of what OwnTracks does.

- The gap follows Equation 1 within a factor of 1.25 across eight speed bands on the two 50 m handsets, with I and D taken from the app, not fitted.
- Changing D to 200 m moves the measured step to 204 m and the crossover to 12 km/h, both where the law puts them, so the relation predicts a configuration it was not fitted on. Below 30 km/h that handset reports two to three times faster than the law allows, which bounds where the displacement trigger is the binding one.
- Spatial resolution is therefore constant and speed-independent, while the packet rate $\max(1/I, v/D)$ swings 33-fold: 2,000 reports per hour at 100 km/h against 60 at rest, invisible to whoever sized the hub.

Related work. Energy-aware sensing makes sampling adaptive and assumes the application owns the policy: EnTracked [1] schedules fixes from a motion model and an error bound, RAPS [2] duty-cycles

on accelerometer, cell-tower and Bluetooth context. Here two rows in a third-party app fix it, the cloud operator never sees them, and the irregularity arrives unexplained. Mode inference [3, 4] and map matching under sparse sampling [5] consume coordinates and treat the period as an exogenous nuisance; under a displacement trigger it is neither. Traffic analysis reads content off the shape of encrypted traffic, for instance the packet *lengths* a variable-bit-rate codec leaks [6]; we read the other observable, arrival time.

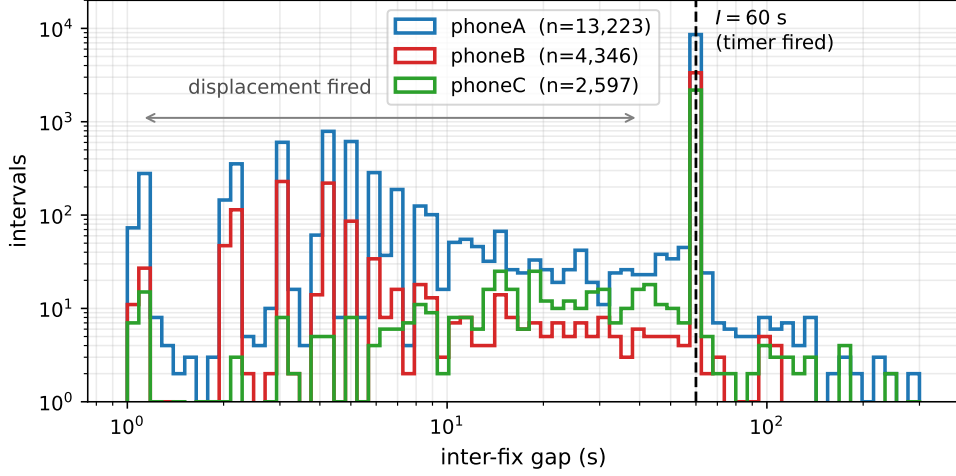


Figure 1: Every inter-fix gap in the capture. Two populations: a spike at the timer and a broad tail below it from the displacement trigger. The comb at 1–5 s is the integer-second resolution of packet arrival.

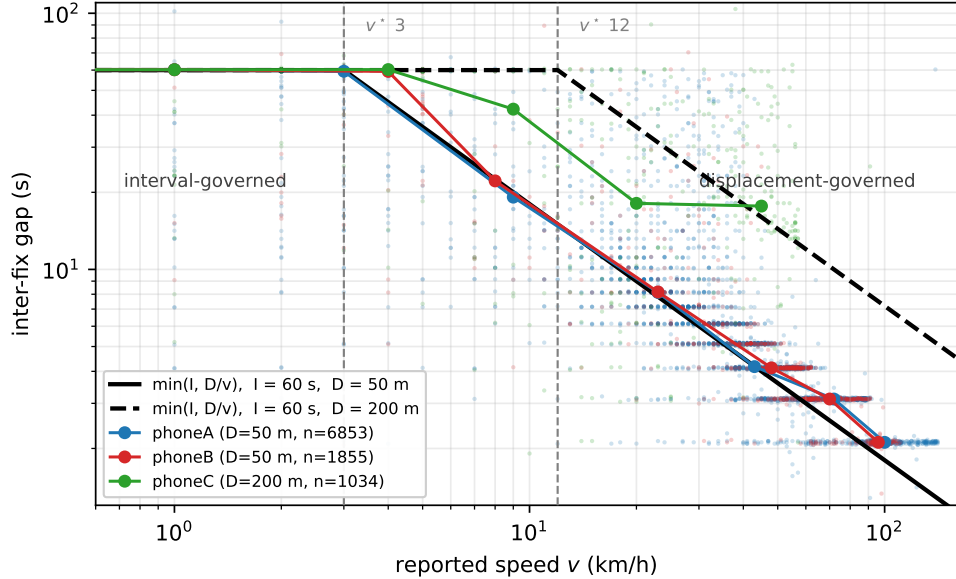


Figure 2: Inter-fix gap against reported speed. The line is Equation 1 with the configured I and D ; it is not a fit. Band ratios of measured to predicted run 0.96–1.25 on phoneA and 0.98–1.32 on phoneB. phoneC, configured at $D = 200$ m, holds 1.10 in its fastest band but falls to 0.50 below 30 km/h.

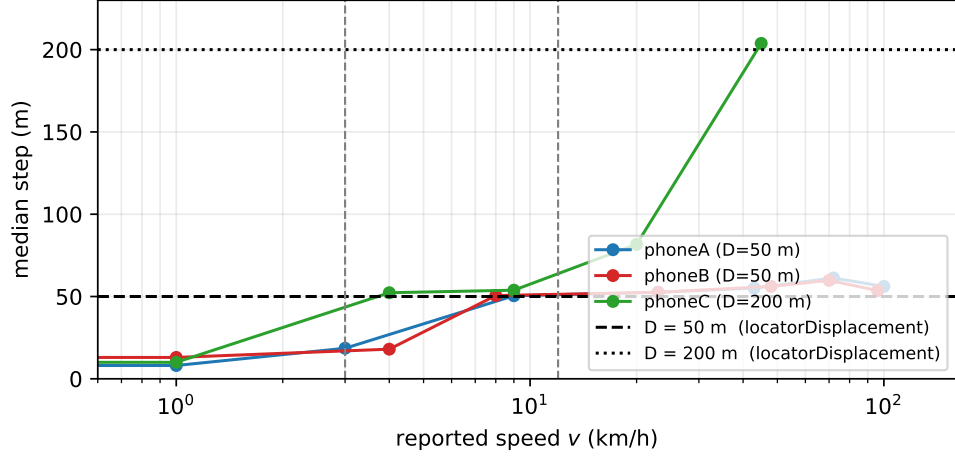


Figure 3: The same fact as distance. Above v^* the step pins near D while the gap varies 30-fold. The overshoot is the trigger firing on the first update to *exceed* D : fitting $D_{\text{eff}} = D_0 + v\tau$ gives $D_0 = 52.6$ m, $\tau = 0.22$ s on phoneA and 54.0 m, 0.09 s on phoneB. phoneC pins at 204 m against its configured 200 m.

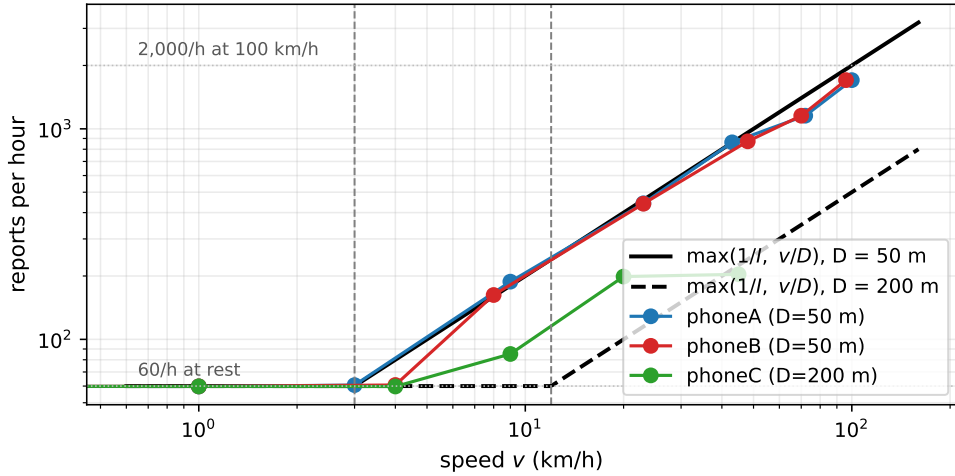


Figure 4: What the law costs whoever sized the hub. Ingest per device is $\max(1/I, v/D)$ and is set by the user's speed, not by the fleet.

2 Inverting it

Equation 1 is invertible on its displacement branch, so a receiver seeing only arrival instants can form $\hat{v}_i = D/(t_i - t_{i-1})$ for gaps under I , reading no field of the payload. Gaps at I are censored: they say only $v \leq v^*$, which is 65.3% of phoneA's intervals, 77.3% of phoneB's, and 84.8% of phoneC's, whose larger D raises v^* to 12 km/h and so censors most of its capture. The reference is not ground truth but the best estimator from the *same* instants that also reads coordinates, $\hat{v}_i^c = s_i/(t_i - t_{i-1})$, because their ratio is an identity,

$$\hat{v}_i/\hat{v}_i^c = D/s_i, \quad (2)$$

so the whole cost of discarding coordinates is the spread of the step about D and nothing else. Table 1 reports both. Three classes follow with one free threshold: a gap at I is a **stop**; a shorter gap splits at $V_W = 12$ km/h into **walk** and **drive**. Figures 5–12 are the estimator, its error, the negative result, the segmenter and the map.

Table 1: Median relative error against reported speed, and segmentation accuracy. The coordinate-reading estimator is the denominator: it shares the sampling instants and adds the positions. phoneC is the same estimator on a handset configured for a four times longer step; it is reported to show what the configuration costs, not as a third replicate.

	coordinates read	timing only, configured D	timing only, D_{eff}	3-class accuracy
phoneA	6.1%	17.5%	15.0%	90.2% ($n = 6,853$)
phoneB	5.9%	18.1%	13.3%	94.0% ($n = 1,855$)
phoneC ($D = 200$ m)	18.4%	59.2%	—	89.2% ($n = 1,035$)

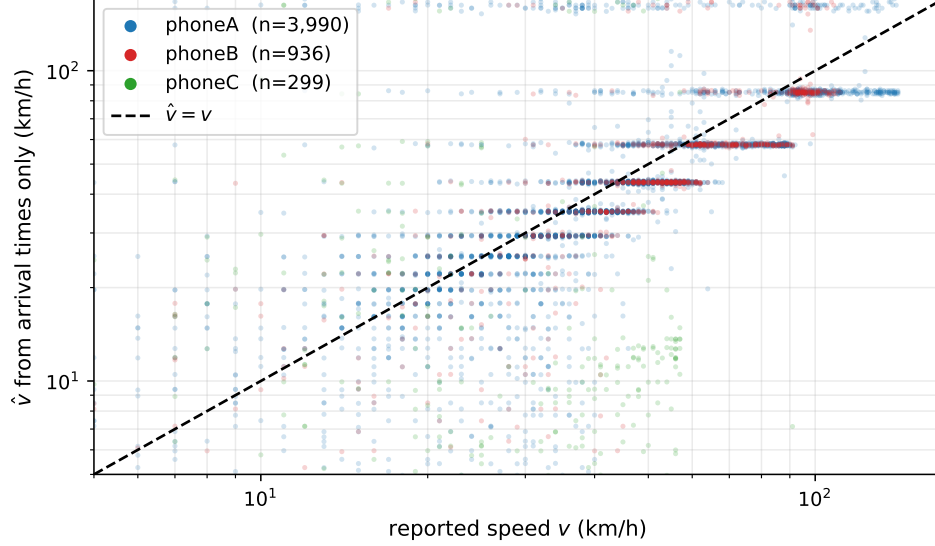


Figure 5: Speed recovered from arrival times alone, against CoreLocation’s reported speed. The horizontal banding is the same integer-second quantisation visible in Figure 1, mapped through $\hat{v} = D/g$.

Setup. Three iPhones, OwnTracks 26.2.3, one device token each, over HTTPS to ThingsBoard CE on EC2; 2026-09-22 to 09-30; 13,252 fixes over 192.4 h, 4,348 over 72.7 h and 2,601 over 47.5 h; 516 km; 23 keys per fix. Two handsets ran $D = 50$ m and the third $D = 200$ m, the only configured difference between them. Intervals over 300 s are the app suspended, not the sampler, and are excluded; one blackout ran 15.9 h under iOS Low Power Mode (Figure 13). Grading is restricted to $v \geq 5$ km/h, below which one count of integer vel exceeds 20%.

Limitations. The sweep of D is two points, not a curve, and it succeeds only above 30 km/h: phoneC’s fastest band matches the law to 1.10 while its 15–30 km/h band reports twice as often as $200/v$ allows, and its 5–15 km/h band 1.4 times. Something other than the displacement filter is releasing fixes at low speed, and with $D = 50$ m that regime is too narrow to have shown up. I was never varied at all. Ground truth is vel from the subsystem that also fires the trigger, so common-mode error would not show. Three iPhones of one vendor and one app version, and the estimator is censored on most intervals, silent about rest. The co-location detector of Figure 11 resolves one pair and not the other: phoneA and phoneC clear the 25 m threshold nine times for 68 minutes in total on 09-30, closest approach 0.4 m with the two handsets on different campus networks, while phoneA and phoneB never come within 4.6 km across 72.7 h and 4,186 jointly fresh cells. The positive case is one afternoon on one pair, so it demonstrates the detector rather than measuring how often people meet. Battery did not separate by motion class in aggregate, 4.9 against 4.7%/h on phoneA, contradicting 3.8 and 11%/h from single windows; we report the disagreement rather than either number.

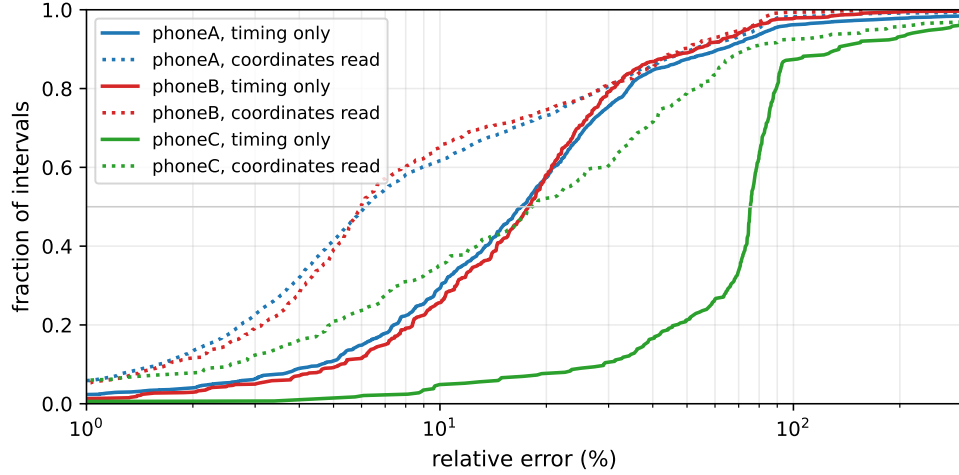


Figure 6: The whole error distribution, not just its median. Discarding the coordinates costs roughly a factor of two to three at every quantile, and by Equation 2 that cost is exactly the spread of D/s_i .

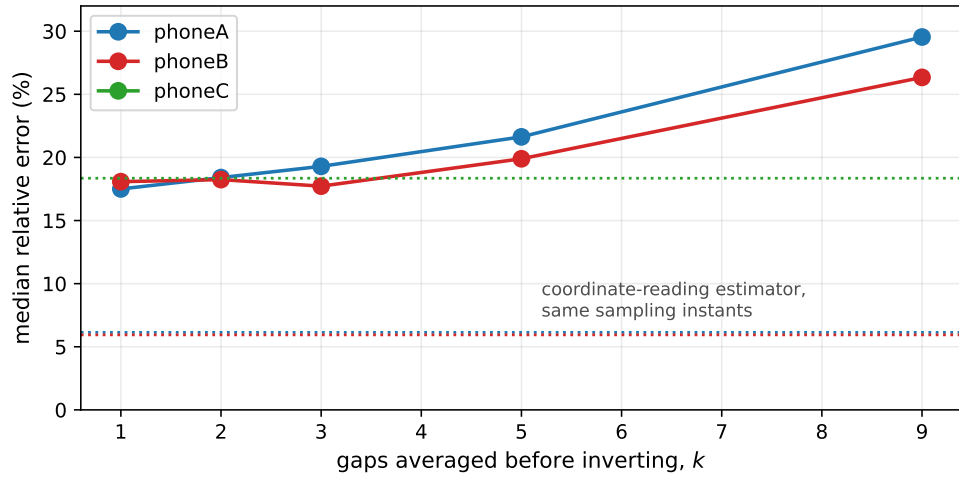


Figure 7: Averaging consecutive gaps makes the estimate worse, monotonically. Independent per-sample noise would fall as $\sqrt{k}$, so the error is dominated by speed that genuinely differs between one 50 m segment and the next, and the estimator is already as local as it should be.

References

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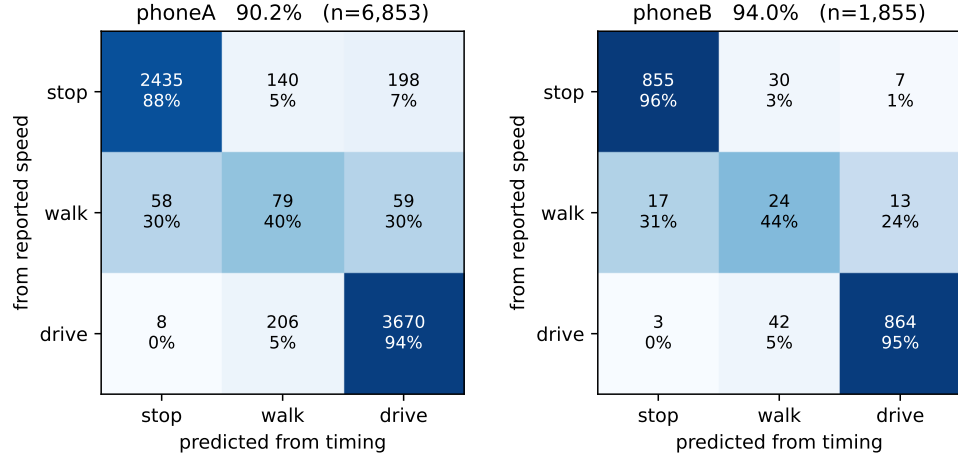


Figure 8: Where the segmenter is wrong, and it is one class. Walk straddles v^* , the one speed at which the two triggers compete; stop and drive are recovered above 92%.

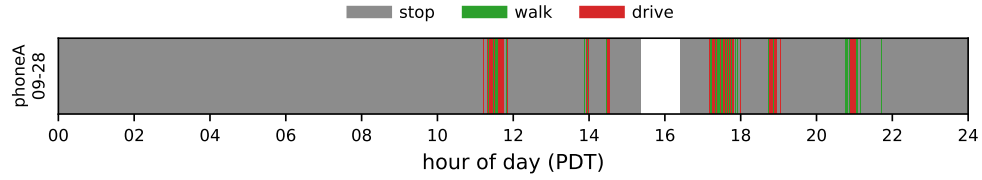


Figure 9: One full day, segmented from arrival times alone. The overnight block is the interval branch saturating; the afternoon is the displacement branch resolving trips and the walks at either end of them.

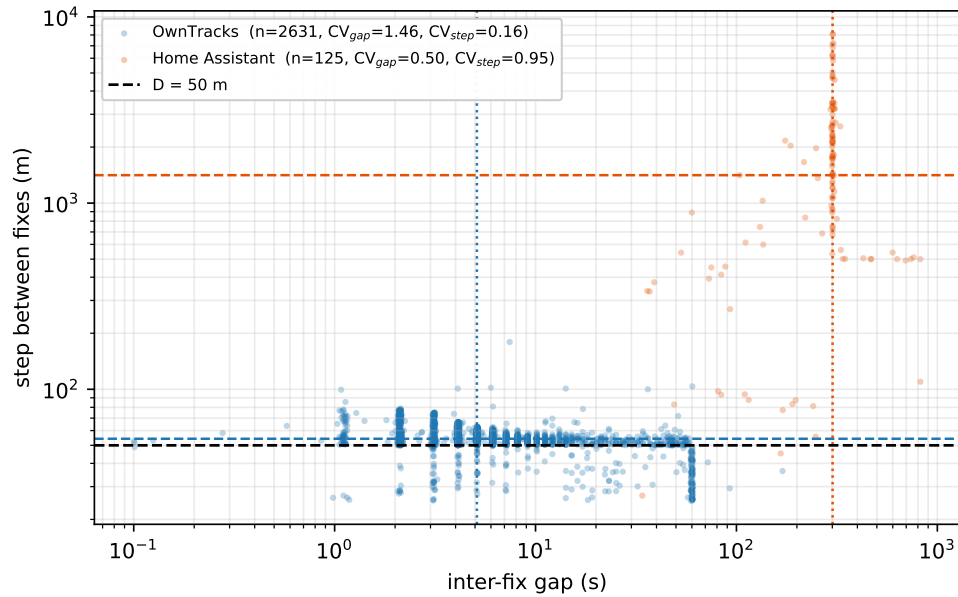


Figure 10: The same handset publishing through two apps over the same hours. Each point is one consecutive pair. OwnTracks holds the step and lets the gap run three decades; the Home Assistant companion holds the gap and lets the step run three decades.

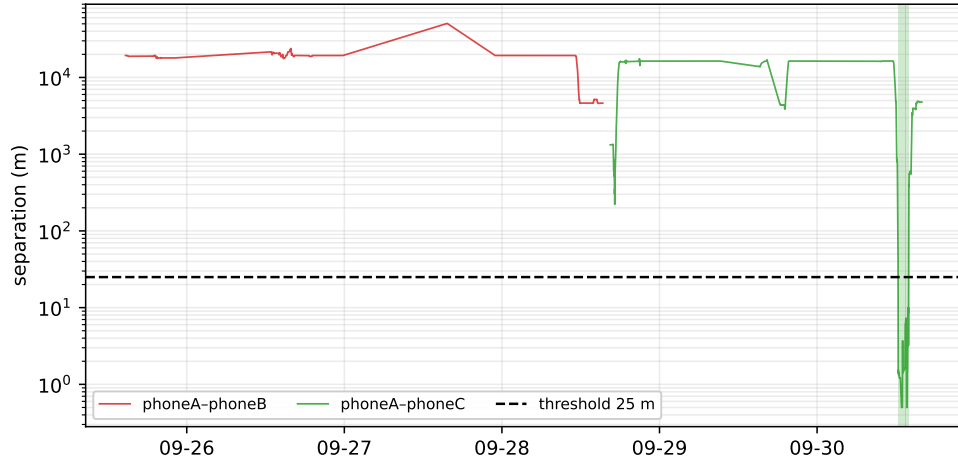


Figure 11: Separation between each pair on a common 30 s grid, with fixes older than 180 s dropped rather than carried. Shaded spans are the intervals that clear the threshold after both accuracy figures are subtracted.

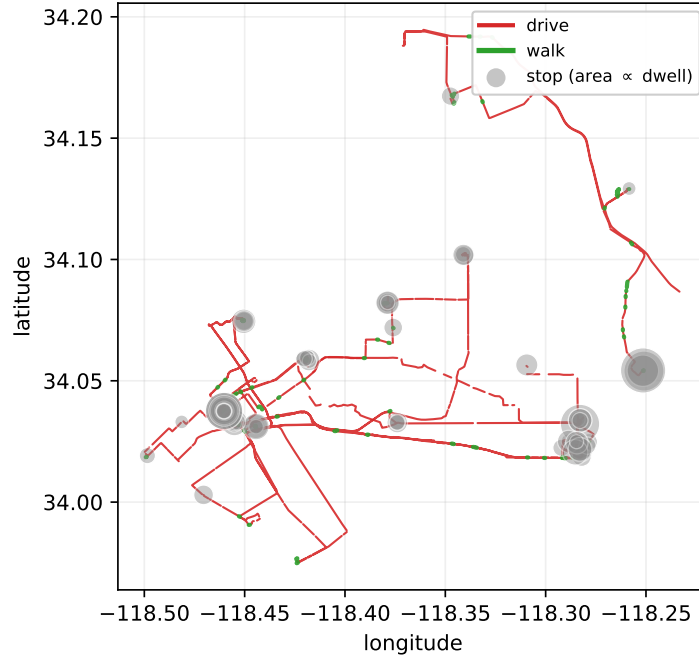


Figure 12: Both handsets over the whole capture, classified from timing only. Stops are circles with area proportional to dwell, since a stop has no spatial extent. The road network is traced entirely by the displacement regime. Interactive: out/mode_map.html.

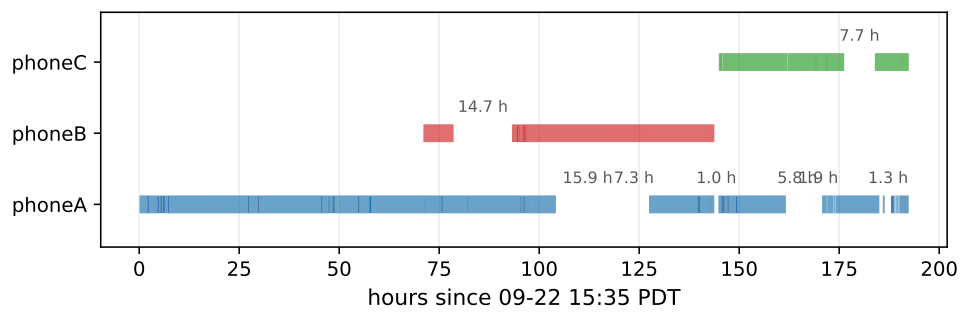


Figure 13: When each phone was actually reporting. The gaps are the app suspended, not the sampler idling; the 15.9 h one is iOS Low Power Mode below 20% battery. Intervals spanning them are excluded throughout.